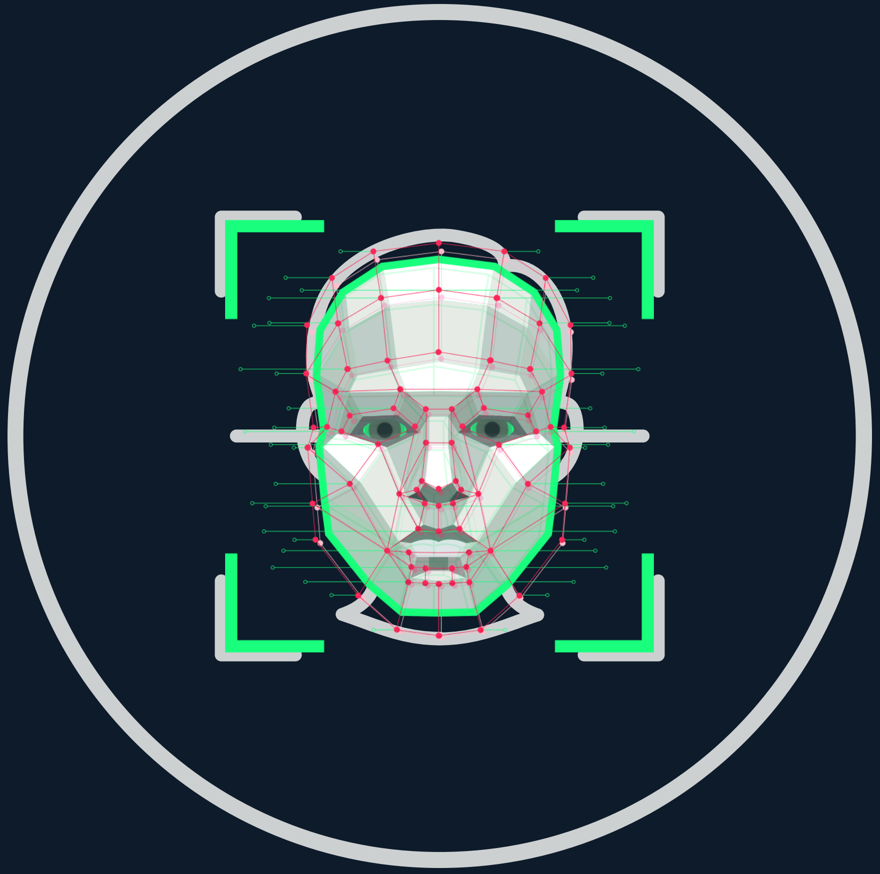

VISIONX

Authenticate Offline · Sync Automatically

Offline Liveness Detection

Edge AI Authentication

Auto Background Sync

First Edge-AI Facial Authentication System for Datalake 3.0

TEAM MAHAVEERA · NHAI Hackathon ·

PULKIT VERMA **ANUPRIYA SAXENA** **ISHITA SAXENA** **SHREYASVI SHRIVASTAVA**

THE PROBLEM: CONNECTIVITY DEAD ZONES

Cloud Dependency

100% reliance on internet — systems completely fail in remote areas, mines, and rural sites.

No Real Liveness Detection

Basic offline apps spoofed by photos or printed masks — zero anti-spoof protection.

3–5 Second Auth Delays

Cloud round-trips destroy field productivity; golden workflow breaks on network timeouts.

Data Loss on Connectivity Drop

Attendance records lost mid-session with no automated sync or recovery mechanism.

Biometric Security Gaps

Local storage without encryption exposes sensitive biometric data to theft or misuse.

Hardware Cost Barrier

Existing biometric scanners cost 5–10× more than a standard Android field device.

"Authentication failures are not just UX issues — they are operational security failures costing data, time, and trust."

SOLUTION & SYSTEM ARCHITECTURE

Camera Feed
Live front camera

▶

Liveness Agent
BlazeFace + MediaPipe

▶

Embedding Agent
MobileFaceNet 128-d INT8

▶

Matching Agent
Cosine vs SQLite DB

▶

Sync Agent
FIFO queue ACK-purge

Auth ✓

LIVENESS SCORE = Blink EAR (0.40) + Head Yaw (0.40) + Depth Variance (0.20) · CRITICAL ≥ 0.80 | HIGH ≥ 0.60 | MEDIUM ≥ 0.40

MODEL FOOTPRINT

BlazeFace
< 1.5 MB
~200 fps · Detection

MediaPipe FaceMesh
~4 MB
468 pts · Liveness

MobileFaceNet INT8
~9 MB
128-d · Embedding

TOTAL
~14.5 MB
< 1 sec · 5.5 MB spare

⚡ Datalake 3.0 Integration: React Native + JSI native modules (Kotlin) · VisionCamera live feed · SQLite FIFO auto-sync · Drops into existing codebase in days

Night Vision: CLAHE pre-processing for low-light & harsh sunlight · Auto PDF attendance reports · GDPR: auto-purge after sync ACK

INNOVATION

Edge AI Model Efficiency

- MobileFaceNet — ArcFace trained, Asian/SA optimised 128-d embeddings
- BlazeFace detector < 1.5 MB at ~200 fps on Android front camera
- MediaPipe 468 facial landmarks for precise geometric liveness
- TensorFlow Lite runtime — CPU-only, no GPU/NPU required
- Sub-1 second full pipeline on Snapdragon 665+ (mid-range)

Compression Techniques

- INT8 quantisation on MobileFaceNet: 9 MB vs 36 MB FP32 baseline
- Total model stack ~14.5 MB — 5.5 MB headroom under 20 MB limit
- Knowledge distillation from full ResNet-50 ArcFace teacher model
- ONNX → TFLite conversion pipeline for portable deployment
- Single-pass inference — no iterative refinement loops

Offline Liveness Detection

- 9-parameter confidence engine: Blink EAR, Head Yaw, Depth Variance, Texture Anti-Spoof, Landmark Stability, Pupil Distance, Face Symmetry, Motion Delta, Light Consistency
- Purely geometric — no second neural network for liveness
- 4-tier confidence scoring: CRITICAL ≥ 0.80 / HIGH ≥ 0.60 / MEDIUM ≥ 0.40 / LOW ≥ 0.20
- Works entirely on-device — zero server call needed

INNOVATION : Unique 9-param geometric liveness (no 2nd net) · INT8 quantised at 14.5 MB · Sub-1s on mid-range Android

FEASIBILITY & INTEGRATION

DATALAKE 3.0 INTEGRATION

React Native + JSI Bridge

Native Kotlin modules via JSI — bypasses slow JS bridge, achieves C++ execution speed for camera & inference.

VisionCamera Integration

Live frame processor hooks feed directly into BlazeFace — no intermediate encoding/decoding overhead.

SQLite + Datalake 3.0 API

Local SQLite DB mirrors Datalake 3.0 schema; sync worker uses existing REST endpoints — zero new API required.

Encrypted Enrollment DB

AES-256 encrypted SQLite stores 128-d embeddings; enrollment UI uses existing role-based access (Admin/User).

Drop-in in Days

Modular npm package — add to existing RN project, configure API base URL, register native modules. No infra changes.

MID-RANGE DEVICE PERFORMANCE

< 1 sec
End-to-end auth pipeline
(Snapdragon 665+)

~200 fps
BlazeFace detection
(front camera feed)

~14.5 MB
Total model footprint
(20 MB limit)

Android 8+
Minimum OS required
(no special hardware)

Policy & Compliance

- Aligns with Digital India & UIDAI biometric standards
- Compatible with DPDP Act 2023 privacy norms
- Supports Aadhaar offline XML-based auth flow
- CPU-only TFLite — open source, zero vendor lock-in

FEASIBILITY : Drop-in RN module · JSI C++ bridge · Sub-1s on Snapdragon 665 · DPDP + UIDAI compliant

SCALABILITY & SUSTAINABILITY

⭐ 20 MARKS

OFFLINE → ONLINE SYNC / PURGE

1. **Queue Locally**
   Every auth event written to SQLite FIFO queue immediately — no data ever lost on network drop.

2. **Background Worker**
   Android WorkManager retries sync in background — exponential backoff on failures.

3. **ACK Before Purge**
   Local record purged ONLY after HTTP 200 ACK from Datalake 3.0 server — zero data loss guarantee.

4. **P2P Mesh (Phase 2)**
   Inter-device peer sync for extreme dead zones — data hops device-to-device until it reaches connectivity.

5. **Conflict Resolution**
   Timestamp + device ID based conflict resolution; server is source of truth post-sync.

LIGHTING & DEMOGRAPHIC ADAPTABILITY

Night Vision — CLAHE
Contrast Limited Adaptive Histogram Equalisation pre-processes every frame for low-light and harsh sunlight — no extra model.

Asian/SA Demographic Optimisation
MobileFaceNet trained with ArcFace on diverse South Asian & Asian datasets for reduced false-rejection rates.

Federated AI (Phase 3)
Each deployed device contributes local gradients to improve the global model — accuracy improves with scale without sharing raw data.

Robust Geometry — Any Face Angle
MediaPipe 468-landmark mesh handles head tilts, partial occlusion, and glasses reliably.

Modular Agent Architecture
Swap any agent (liveness, embedding, matching) independently for regional or regulatory customisation.

SCALABILITY : ACK-before-purge SQLite FIFO · CLAHE night vision · Federated AI roadmap · Zero data loss guarantee

DOCUMENTATION, MARKET & ROADMAP

Innovation
Edge AI + Liveness

Feasibility
RN Integration

Scalability
Sync + Diversity

Presentation
Docs + Clarity

DOCUMENTATION & CODE QUALITY

Clean Modular Source Code
5-agent pipeline separated into distinct modules (liveness, embedding, matching, sync, UI)

Integration Guide
Step-by-step RN setup doc: install, configure, enroll, authenticate, sync — with code snippets

Test Suite & Benchmarks
Unit tests per agent; benchmark results on Snapdragon 665, 680, and 720G devices

Security Architecture Doc
Threat model, AES-256 key management, GDPR purge policy, DPDP Act compliance notes

ROADMAP & MARKET

① NOW — DatalakeFaceAuth
Sub-1s offline auth · 9-param liveness · SQLite FIFO · VisionCamera RN

② 12 MO — Datalake Intel
Auth-failure heatmaps · P2P mesh sync · Web portal · HR module auto-sync

③ LONG-TERM — Datalake Chain
Blockchain identity logs · Federated AI · National referral tracking

MARKET: ₹2,400 Cr field auth (India/yr) · 120M+ Android field devices · USD 68B biometric market · Goal: default auth layer for Datalake 3.0 worldwide